

Invert Lab

DOTS Sprite Animator

User Guide

Version 0.8.0

Package: com.invertlab.spriteanimator

Namespace: InvertLab.Sprites.DOTS

Unity 6000.0+ · Entities · Entities Graphics · URP

Portable DOTS-first 2D flipbook animation: timeline authoring, events, colliders, sockets, combat playback helpers, and CPU/GPU paths.

1. Product overview

Invert Lab DOTS Sprite Animator is a portable Unity package for DOTS-first 2D flipbook authoring and runtime playback. Publisher branding is **Invert Lab** (not BallForge). Runtime namespace: **InvertLab.Sprites.DOTS**.

Version **0.8.0** adds interrupt/priority playback, queue/one-shot helpers, hitstop/hold/combo facing helpers, ReverseOnce wrap, editor accordion/FPS/event UX, Show Sprite scene preview, sockets + collider bake, and sample scenes for playback, events, colliders, sockets, and crowd GPU.

2. Install and menus

Embed Packages/com.invertlab.spriteanimator. Requires Entities, Entities Graphics, and URP. Run **Tools > DOTS Sprite Animator > Validate Installation** after import.

Menu	Purpose
Window > DOTS Sprite Animator	Main timeline / sheet authoring
Tools > ... > Open Window	Same window
Tools > ... > Validate Installation	Dependency check
Tools > ... > Help	QuickStart / README
Tools > ... > Build Sockets Sample	Rebuild sockets sample
Tools > ... > Build Events Sample	Rebuild events sample
Tools > ... > Build Playback API Sample	Rebuild playback API sample
Tools > ... > Open Crowd GPU Sample	Open crowd/GPU scene
Tools > ... > Setup Authoring Example Scene	Authoring scene helper

3. Quick authoring

1. Open the window → New Profile → assign sheet, rows, columns.
2. Edit clips (accordion), frames, FPS, events, colliders, sockets.
3. Save Profile → asset + JSON.
4. Add SpriteAnimSetAuthoring, assign Profile, bake.
5. Optional SpriteAnimPlayerAuthoring for managed Play / Hitstop / queue APIs.

```
SpriteAnims.Play(em, entity, "Run");  
SpriteAnims.PlayFacing(em, entity, "Walk", SpriteFacingDirection.Down);
```

4. Playback API

4.1 Interrupt and Priority

Field	Meaning
Interrupt Always	Any Play may replace (idle/walk).
Interrupt Never	Locked until Once/ReverseOnce ends, Stop, or force.
Interrupt AfterTime	Cancelable when normalized time ≥ CancelAfter (0–1).
Priority	Higher/equal priority may interrupt when force is false.

Combo windows lower effective current priority via ComboWindowPriorityBoost so follow-ups can land in-window.

4.2 Queue, one-shot, chaining

API	Behavior
PlayOrQueue	Play if allowed; else queue until current finishes.
PlayOneShot	Play once, then resume previous locomotion/idle.
OnCompleteClipIndex	Auto-start this clip when Once/ReverseOnce ends (-1 = none).

4.3 Speed, seek, freeze, blend

API	Behavior
SetSpeed / GetSpeed	Rate; negative = rewind; 0 = freeze clock.
Pause / Resume / Freeze / Unfreeze	Pause without losing restore intent; Freeze aliases Pause.
SeekFrame / SeekNormalized / SetTime	Jump within current clip.
Blend	Crossfade weight 1→0 during fade (shader/gameplay readable).

4.4 Lifecycle

ClipStarted / **ClipCompleted** on SpriteAnimPlayerAuthoring and SpriteAnimEvents (reserved Ids 250/251). Once + negative speed completes at phase 0.

5. Combat helpers

API	Behavior
Hitstop(seconds)	Freeze clock, then restore speed (shared timer with Hold).
Hold / HoldAtFrame	Freeze; HoldAtFrame seeks then holds.
Combo window	Start/End frame + PriorityBoost; InComboWindow / TryComboPlay.
SetFacing / PlayMirrored / flipX Play	Facing and mirror helpers.
PlayRandomStart	Play + seek random frame (crowd variety).
PlayWeighted	Weighted random among clips (arrays or up to 4 pairs).

6. Wrap modes

Mode	Behavior
Loop	Repeats forward.
Once	Forward once, then completes.
Ping Pong	Forward then reverse, repeating.
Reverse Loop	Loops while playing backward.
ReverseOnce (0.8)	Once from last→first with positive Speed; then completes / chains.

Ping-pong, reverse, custom holds, events, sockets, or TRS tween → **CPU only**.

7. Editor UX (0.8)

- Clip accordion · FPS edits · EVENT TYPES multi-select/delete · event-marker RMB menu · rename UX fixes · frame-delete persists on Save Profile.
- **Show Sprite** scene preview (was Show Scene Preview; ShowSpriteInScene).
- Sockets click-to-place, inventories, Independent Motion + triggers.
- Unity collider bake/gizmos: Square/Circle/Polygon; Frame/Clip/Character lifetime; Query AABB / Unity 2D / Both.

8. Sockets, independent motion, inventory

Socket **Name** is a label; **ID** (e.g. equipment.head, combat.muzzle) is the stable contract. Frame-attached keys follow the flipbook; Independent Motion has its own timeline and RMB triggers (SpriteSocketEventBuffer).

SyncUnitySockets keeps GO transforms aligned. Attach with SpriteSocketAttachmentAuthoring. Lookup: SpriteSockets.TryGetWorldPose / TryGetPose. Sockets force CPU playback.

See Documentation~/SocketGameplayAPI.md.

9. Events — Footstep / Attack story

Markers: byte Id, in-frame time, Loop/Once fire mode, typed payloads. Typical story: Footstep on walk contact frames; Attack on impact. Consume via SpriteAnimEventBuffer or SpriteAnimEvents.Raised. EventMarkers is source of truth (multi per frame). Event clips stay on CPU. See AnimationEvents.md.

10. Crowd / GPU overview

Path	When
CPU (SpriteAnimPlayerSystem)	Events, sockets, holds, reverse/ping-pong, combat markers.
GPU clock	Simple uniform Loop/Once; badge GPU clock OK.

SpriteGpuAnimSwitch.ToGpu refuses non-eligible clips. Crowd: SpriteCrowdSpawnerAuthoring primes instance scale. Sample: Assets/Samples/CrowdGpuExample. Preview mesh uses SpriteUnlit2DPreview.

11. Samples

Folder	Shows	Open via
ColliderEventExample	Colliders + event reactions	Scenes/ under sample
EventsExample	Footstep/Attack + ClipStarted/Completed	Tools > Build Events Sample
PlaybackApiExample	Play / OneShot / Queue / Priority / Hitstop / Hold	Tools > Build Playback API Sample
CrowdGpuExample	Dense GPU crowd path	Tools > Open Crowd GPU Sample
Sockets	Weapon socket Idle/Attack	Tools > Build Sockets Sample
Showcase/Clembod	Warrior / Bringer demo art	License + Contact (credit appreciated)

Showcase art © Clembod — personal/commercial OK; do not redistribute/resell art alone. Credit appreciated: @Clembod / clembod.itch.io.

12. Related package docs

[QuickStart.md](#) · [AnimationEvents.md](#) · [SocketGameplayAPI.md](#) · [Documentation.md \(this source\)](#) · [CHANGELOG.md](#) · [README.md](#)

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